

Exploratory Data Analysis

120 Years of Olympic History Dataset

Python Data Analysis Project | Pandas, NumPy, Matplotlib & Seaborn

AMAYA SUDHAKAR

SECTION 01

Introduction

Project context, analytical scope, and historical
relevance

Project Overview & Background

Analysis Context

Exploratory Data Analysis (EDA) is a crucial step in any data analysis workflow. This project dives into modern Olympic history, spanning over a century of global athletic excellence.

Using Python, we parse complex multi-variable data to reveal trends in athlete demographics, country performances, gender participation, and event evolutions.



SECTION 02

Data Source Overview

Data origin, dataset attributes, and structural
breakdown

Data Schema & Column Description

Attribute Name	Data Type	Description	Example Value
ID / Name	Integer / String	Unique athlete identification number and name	10630 / Ole Einar Bjørndalen
Sex / Age	Categorical / Numeric	Gender indicator (M/F) and athlete's age	M / 27.0
Height / Weight	Float (Numeric)	Physical attributes in cm and kg	184.0 cm / 80.0 kg
Team / NOC	Categorical	Represented country and National Olympic Committee code	Norway / NOR
Games / Year	String / Integer	Olympic edition year and season (Summer/Winter)	2002 Winter / 2002
Sport / Event / Medal	Categorical	Specific discipline, competition event, and award	Biathlon / Sprint 10km / Gold

SECTION 03

A Glimpse on Data Source

Data volume, profiling, cleaning, and technical
setup

Data Profiling & Preprocessing



Massive Dataset Scale

Contains 271,116 records covering modern Olympic history from Athens 1896 to Rio 2016 across both Summer and Winter games.



Data Quality & Cleaning

Handled missing values in physical attributes (Height, Weight, Age) and imputed missing Medal fields with 'No Medal' categorizations.



Python Stack

Utilized Pandas for data wrangling, NumPy for numeric operations, and Seaborn / Matplotlib for rich visual exploratory plots.

SECTION 04

Project Objectives

Core research questions and analytical
targets

Core Analytical Objectives

Performance & Demographics

- Identify overall top medal-winning nations across modern games history.
- Evaluate the growth of female athlete participation over the decades.
- Analyze physical attributes (Height vs Weight correlation) across different sports disciplines.

Historical Evolution

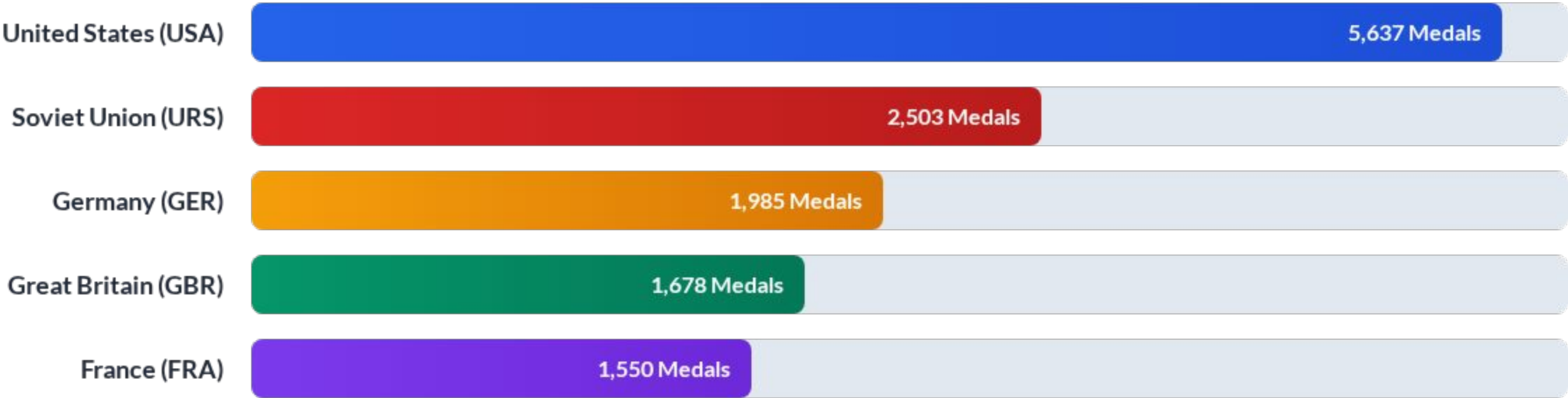
- Track country participation trends and geographical expansion.
- Discover top individual medalists with peak performances.
- Examine age distribution among gold medal winners in Summer vs Winter games.

SECTION 05

Key Insights

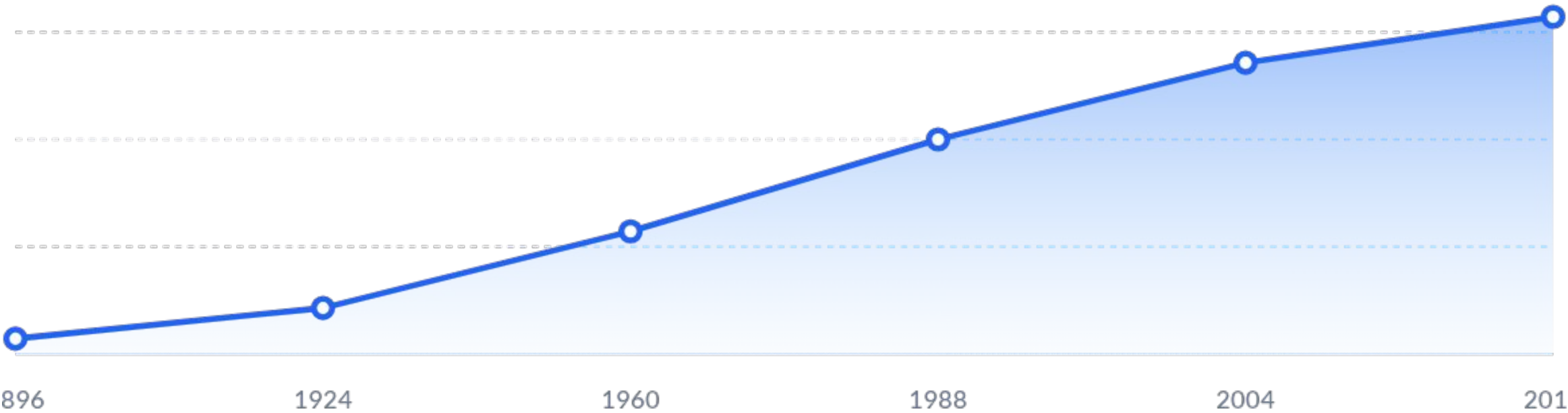
Empirical discoveries, trends, and data
visualizations

Top Medal-Winning Nations



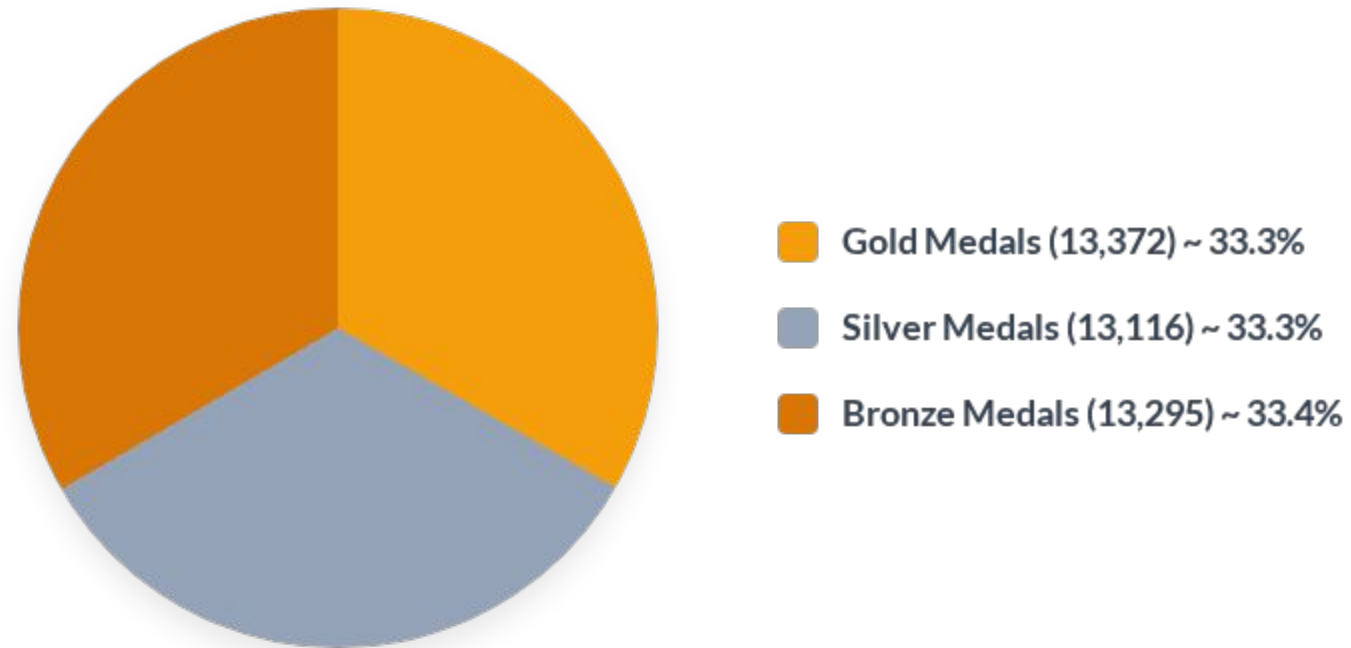
The USA leads historically in overall medal counts, followed by former Soviet Union and Germany, reflecting long-term sporting infrastructure investments.

Athlete Participation Over Time



Continuous exponential growth in total athlete participation, expanding from under 250 competitors in 1896 to over 11,000+ athletes in recent Summer Games.

Distribution of Recorded Medals



Across historical winners, medal distribution remains balanced between Gold, Silver, and Bronze, reflecting standardized podium awarding across sports events.

SECTION 06

Conclusion

Key takeaways, technical achievements, and future research

Conclusion & Key Takeaways



Analytical Insights

Female participation expanded dramatically from ~2% in 1900 to nearly 45% in 2016. Physical attributes vary drastically by sport (e.g., Basketball vs Gymnastics).



Technical Execution

Demonstrated effective Python data manipulation, aggregation, feature engineering, and data visual storytelling using Pandas and Seaborn.



Future Work

Future scope includes predictive modeling for national medal counts using machine learning and building interactive Streamlit web applications.

Questions & Answers

Thank you for exploring the Olympic Data Analysis presentation!

[Python EDA Project | Olympics Dataset Analysis](#)